

DZ MOTOR 大中电机

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<https://www.dzmotortech.com>



DAZHONG MOTOR TECH CO., LTD.

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🌐 <https://www.dzmotortech.com>

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DaZhong Miniature Servo Motor

Semiconductor/Optics/Medical/Humanoid Robot

COMPANY PROFILE

1958

Founded in

1680

members

68%

professional
technicians

340

senior and master
-level technicians



Jiangsu Dazhong Electric Motor Co., Ltd. is a modern intelligent factory integrating R&D, design, manufacturing, sales and service of electric motors. It is also a national-level high-tech enterprise, founded in 1958 with a registered capital of 220 million yuan.

Currently, the company has 2,000 employees, among whom professional and technical personnel account for 68%, including 1,360 technicians and senior technicians. It covers an area of 2 million square meters, with an annual production capacity of up to 1 million units / 14 million kilowatts. Relying on its solid technical strength and large-scale production foundation, the company's core products cover a variety of high-efficiency and energy-saving Servo motors, VFDs, high & low voltage motors, high-efficiency flameproof motors, special motors, permanent magnet motors, new energy motors and other synchronous / asynchronous motors.

These products are widely used in key fields such as power stations, metallurgy, coal mines, machinery, petroleum, chemical industry, shipping, transportation, water conservancy, cement, papermaking and environmental protection. In the industry, the company has steadily ranked among the top three in terms of enterprise scale and comprehensive economic benefits. Among its products, flameproof motors rank high in market share, and the domestic market share of its level-1 energy efficiency motors even reaches 45%.

In terms of core competitiveness building, the company has more than 30 core competitive products and over 30 patents of various types. Among them, the independently developed 15# and 25# flange micro servo motors are the first domestic products that can realize import substitution, successfully filling the market gap of domestic self-developed products in this field. Meanwhile, the company has participated in formulating two industrial Internet industry standards, laying an important foundation for promoting the in-depth integration of intelligent motors and industrial Internet platforms. In addition, the company boasts a strong core team: its founder is from the Chinese Academy of Sciences (CAS), and members of its process automation team all come from international large enterprises.

At present, the company has established cooperation with a number of scientific research institutes and universities, and built a sound industry-university-research cooperation and talent training mechanism, continuously injecting impetus into technological innovation and talent reserve.

In the future, the company will continue to adhere to the development concept of technological innovation, deepen its focus on the field of intelligent manufacturing, facilitate industrial upgrading through intelligent manufacturing, and strive to realize the new-type industrialization development model featuring the integration of informatization and industrialization, so as to continuously lead the high-quality development of the industry.



SYSTEM CERTIFICATION



Product Center



15 Servo Motor



25 Servo Motor



Integrated Ball
Screw Machine



Integrated Speed
Reducer



Integrated Harmonic
Speed Reducer



Electric Push Rod



25E Servo Motor



25E Servo Motor
with Brake



40D Servo Motor



40D Servo Motor
with Brake



Servo Electric Cylinder



Circular Servo (Electric
Screwdriver)

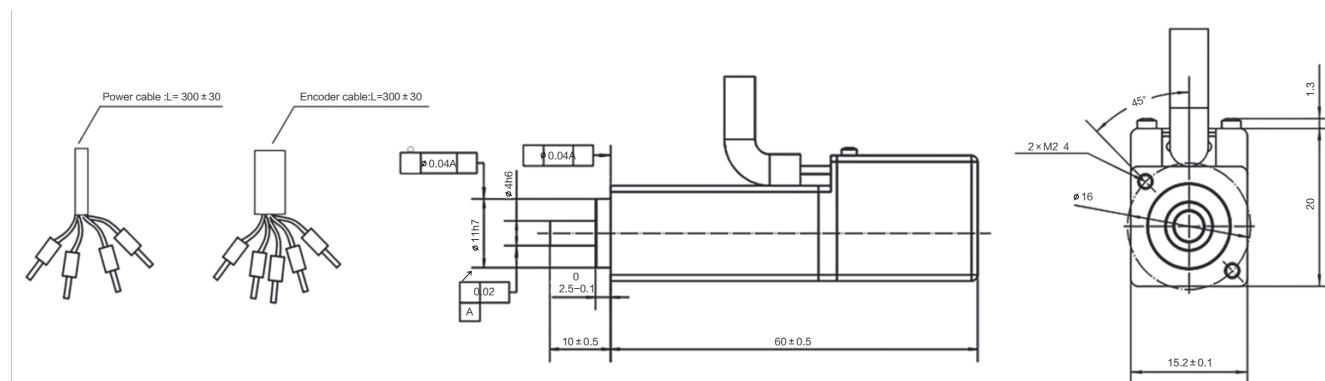
► 15D Servo Motor

5W 48V

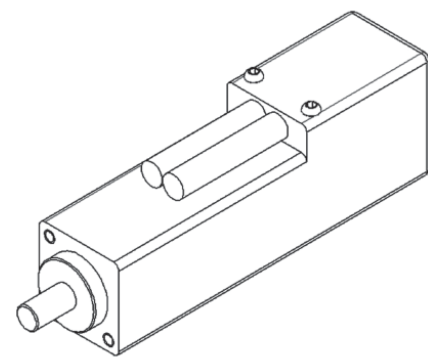
- ◎ Maximum speed: 6000rpm
- ◎ Waterproof grade: IP50
- ◎ Peak torque: 0.048Nm
- ◎ Rated power: 5W



Product Appearance



Product Specifications



Pin Number	Signal	Wire Color
1	PE	Black
2	5V	Orange
3	0V	Black
4	SD+	Green
5	SD–	Yellow

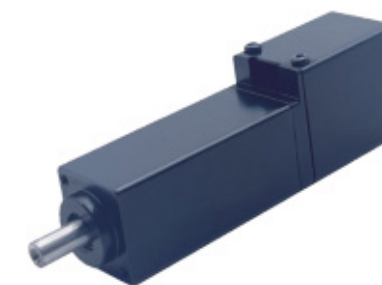
Pin Number	Power cord definition	
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow-green

Rated Voltage	V	DC48
Rated Power	W	5
Number of Pole Pairs	/	2
Static Torque	Nm	0.016
Rated Torque	Nm	0.016
Peak Torque	Nm	0.048
Static Current	Arms	1.5
Rated Current	Arms	1.5
Peak Current	Arms	4.5
Rated Speed	rpm	3000
Maximum Speed	rpm	6000
Rotational inertia	10 ⁻⁴ kgm ²	0.0007
Torque coefficient	Nm/A	0.011
Protection class	/	IP50
Insulation class	/	Class F
Wire resistance	0 hm	2.18
Wire inductance	mh	0.15
Wire back EMF	V/krpm	0.54
Total length of motor(L)	mm	60

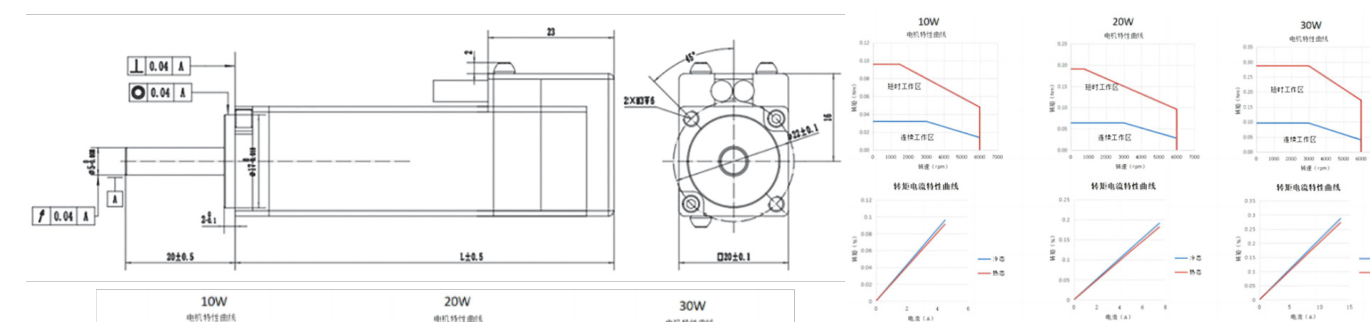
➤ 20 Servo Motor

10~30W

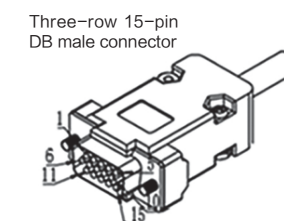
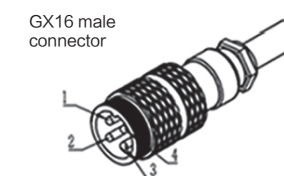
- ◎ **Rated Speed:** 3000rpm
- ◎ **Waterproof Grade:** IP50
- ◎ **Rated Torque:** 0.032/0.064/0.096Nm
- ◎ **Product Appearance:** 10W/20W/30W



Product Appearance



Product Specifications



Power Line Definition		
Phase Sequence	Phase Sequence	Wire Color
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow – green

Wire Color Definition of 17bit Single – turn Absolute Magnetic Encoder		
Pin Number	Signal	Wire Color
1	PE	Black
2	5V	Orange
3	0V	Black
4	SD+	Green
5	SD–	Yellow

Model SY20D00 □ 30M3002-24V	Power W	Rated Torque Nm	Rated Speed rpm	Body Length mm
SY20D00130M3002-24V	10	0.032	3000	69
SY20D00230M3002-24V	20	0.064	3000	89
SY20D00330M3002-24V	30	0.096	3000	109

Motor input voltage	VDC	24V	24V	24V
Servo motor model	/	SY	SY	SY
Flange specification	mm	□ 20	□ 20	□ 20
Rated output	W	10	20	30
Number of pole pairs	/	2	2	2
Rated torque (hot state)	Nm	0.032	0.064	0.096
Instantaneous maximum torque	Nm	0.096	0.191	0.287
Rated current	Arms	1.5	2.5	4.5
Instantaneous maximum current	Arms	4.5	7.5	13.5
Rated speed	rpm	3000	3000	3000
Maximum speed (At maximum speed)	rpm	1500	750	3000
Maximum speed	rpm	4000	4000	4000
Torque coefficient	Nm/Arms	0.02	0.03	0.02
Moment of inertia	10-4kgm2	0.0008	0.0016	0.0024
Protection class	/	IP50	IP50	IP50
Insulation class	/	Class F	Class F	Class F
Winding resistance	0 hm	3.60	2.40	0.96
Winding inductance	mh	0.50	0.40	0.19
Winding back EMF	V/krpm	1.29	1.55	1.29

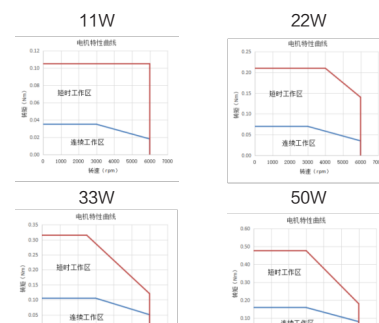
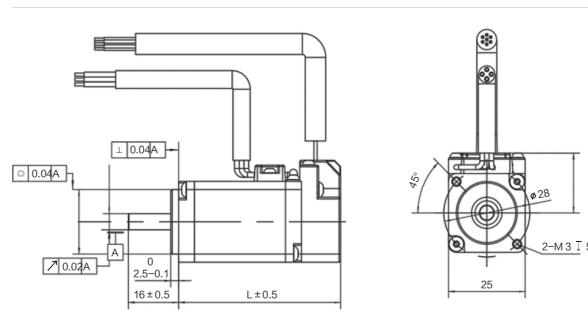
25D Motor

11W/22W/33W/50W 48V

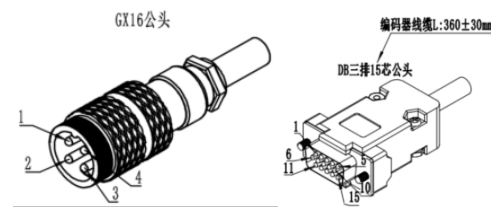
- Maximum speed : 6000rpm
- Protection class : IP50
- Rated torque : 0.035Nm/0.070Nm/0.105Nm/0.159Nm
- Rated power : 11W/22W/33W/50W



Product Appearance



Product Specifications



Power Line Definition		
Pin Number	Phase Sequence	Wire Color
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow – green

Definition of 17bit Absolute Multi – turn Magnetic Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD–	Yellow
6	VB+	Brown
7	VB–	White

Definition of Incremental 2500–line Magnetic Encoder Cable		
Pin Number	Signal	Wire Color
1	GND	Black
2	Vcc	Red
3	W+	Gray
4	V+	Orange
5	U+	Blue
6	Z+	Brown
7	B+	White
8	A+	Yellow
9	/	/
10	/	/
11	/	/
12	/	/
13	Z–	Purple
14	B–	Pink
15	A–	Green

Rated Voltage	V	DC48
Rated Power	W	11W/22W/33W/50W
Number of Pole Pairs	/	5
Static Torque	Nm	0.035/0.070/0.105/0.159
Rated Torque	Nm	0.035/0.070/0.105/0.159
Peak Torque	Nm	0.105/0.210/0.315/0.478
Static Current	Arms	2/2/2/2.6
Rated Current	Arms	2/2/2/2.6
Peak Current	Arms	6/6/6/7.8
Rated speed	rpm	3000
Maximum Speed	rpm	6000
Torque Coefficient	Nm/A	0.0175/0.035/0.0525/0.06
Moment of Inertia	10 ⁻⁷ kgm ²	2.7/4.7/6.7/11.7
Protection Class	/	IP50
Insulation Class	/	Class F
Wire Resistance	Ohm	1.7/2.6/3.5/5.3
Wire Inductance	mh	0.26/0.49/0.73/1.1
Wire Back EMF	V/krpm	1.2/2.4/3.5/4.8
Total Length of Body(L)	mm	51.5/62.5/73.5/90.5

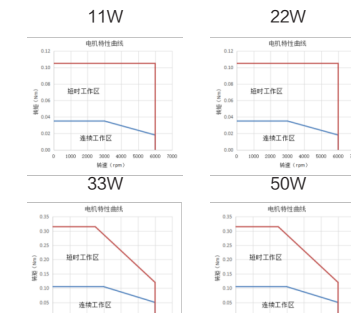
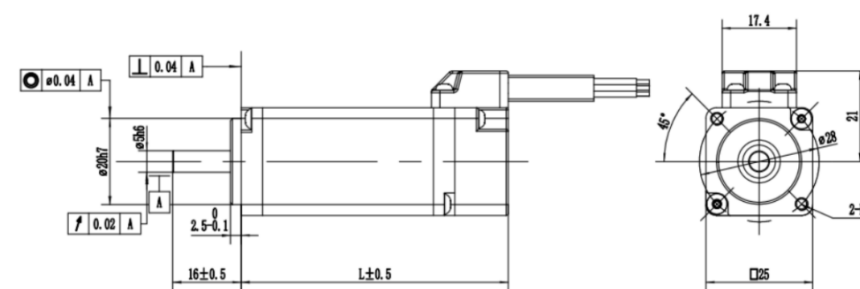
25E Motor

11W/22W/33W/50W 48V

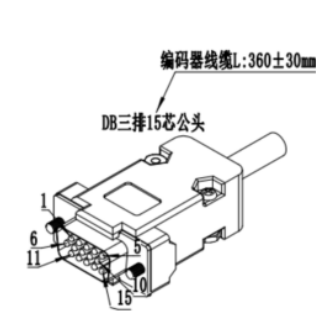
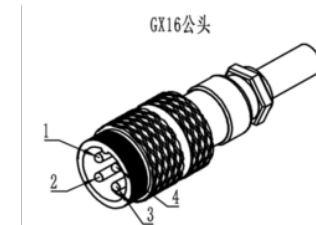
- Maximum speed: 6000rpm
- Protection class: IP50
- Rated torque: 0.035Nm/0.070Nm/0.105Nm/0.159Nm
- Rated power: 11W/22W/33W/50W



Product Appearance



Product Specifications



Power Line Definition		
Phase Sequence	Phase Sequence	Wire Color
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow – green

Definition of 17bit Absolute Multi – turn Magnetic Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD–	Yellow
6	VB+	Brown
7	VB–	White

Rated Voltage	V	DC48
Rated Power	W	11W/22W/33W/50W
Number of Pole Pairs	/	5
Static Torque	Nm	0.035/0.070/0.105/0.159
Rated Torque	Nm	0.035/0.070/0.105/0.159
Peak Torque	Nm	0.105/0.210/0.315/0.478
Static Current	Arms	2/2/2/2.6
Rated current	Arms	2/2/2/2.6
Peak Current	Arms	6/6/6/7.8
Rated Speed	rpm	3000
Maximum Speed	rpm	6000
Torque Coefficient	Nm/A	0.0175/0.035/0.0525/0.06
Moment of Inertia	10 ⁻⁷ kgm ²	2.7/4.7/6.7/11.7
Protection Class	/	IP50
Insulation Class	/	Class F
Wire Inductance	Ohm	1.7/2.6/3.5/5.3
Wire Resistance	mh	0.26/0.49/0.73/1.1
Wire Back EMF	V/krpm	1.2/2.4/3.5/4.8
Total Length of Body(L)	mm	51.5/62.5/73.5/90.5

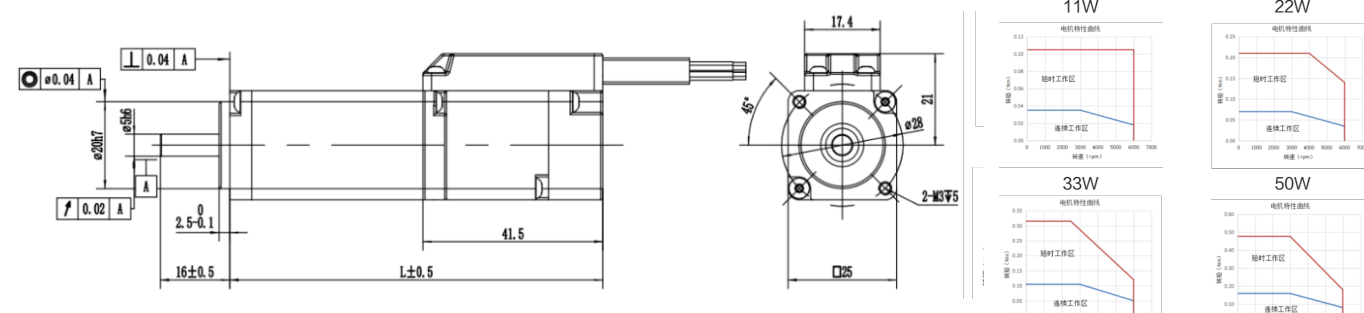
25E Brake Motor

11W/22W/33W/50W 48V

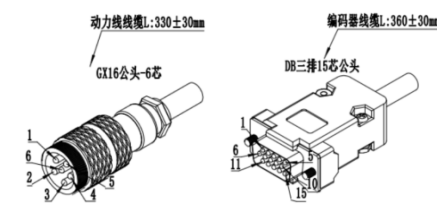
- Maximum speed : 6000rpm
- Protection class : IP50
- Rated torque : 0.035Nm/0.070Nm/0.105Nm/0.159Nm
- Rated power : 11W/22W/33W/50W



Product Appearance



Product Specifications



Brake Parameters	
Attraction/Maintenance Voltage	DC24/ DC7
Attraction/Maintenance Power	16(W)/1.4(W)
Resistance (20°C)	36 ± 7%
Static Torque	≥ 0.02(N.m)
Attraction Time	< 50(ms)
Release Time	< 20(ms)
Release Voltage	0.5~6(V)
Attraction Voltage	< 19.2(V)

Power Line Definition		
Pin Number	Phase Sequence	Wire Color
1	U	Pink
2	V	Orange
3	W	Black
4	PE	Blue
5	Brake	Purple
6	Brake	Brown

Definition of 17bit Absolute Single-turn Magnetic Encoder		
Pin Number	Phase Sequence	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Definition of 17bit Absolute Single-turn Optical Encoder		
Pin Number	Phase Sequence	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

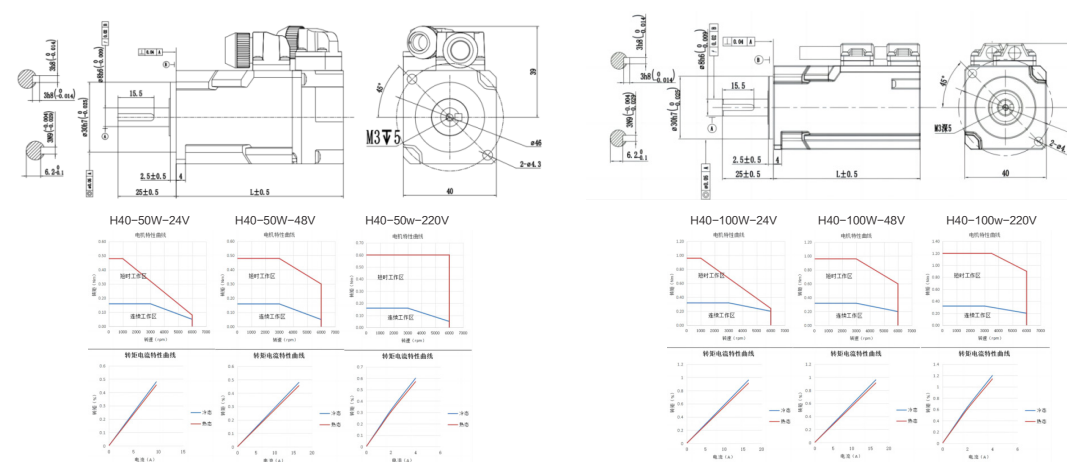
Rated Voltage	V	DC48
Rated Power	W	11W/22W/33W/50W
Number of Pole Pairs	/	5
Static Torque	Nm	0.035/0.070/0.105/0.159
Rated Torque	Nm	0.035/0.070/0.105/0.159
Peak Torque	Nm	0.105/0.210/0.315/0.478
Static Current	Arms	2/2/2/2.6
Rated Current	Arms	2/2/2/2.6
Peak Current	Arms	6/6/6/7.8
Rated speed	rpm	3000
Maximum Speed	rpm	6000
Torque Coefficient	Nm/A	0.0175/0.035/0.0525/0.06
Moment of Inertia	10 ⁻⁷ kgm ²	2.7/4.7/6.7/11.7
Protection Class	/	IP50
Insulation Class	/	Class F
Wire Resistance	Ohm	1.7/2.6/3.5/5.3
Wire Inductance	mh	0.26/0.49/0.73/1.1
Wire Back EMF	V/krpm	1.2/2.4/3.5/4.8
Total Length of Body(L)	mm	75/86/97/114

40D Motor

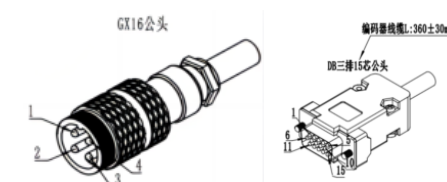
- Maximum speed: 6000rpm
- Protection class: IP54/IP65
- Rated torque: 0.16Nm/0.32Nm
- Rated power: 50W/100W



Product Appearance



Product Specifications



Definition of 17bit Absolute Multi - turn Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Blue
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

23bit Absolute Multi - turn Encoder Definition		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Blue
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Cable Definition of Incremental 2500 - line Magnetic Encoder		
Phase Sequence	Signa	Wire Color
1	GND	Black
2	VCC	Red
3	W+	Gray
4	PE	Orange
5	U+	Blue
6	Z+	Brown
7	B+	White
8	A+	Yellow
9	/	/
10	/	/
11	/	/
12	/	/
13	Z-	Purple
14	B-	Pink
15	A-	Green

Power Cable Definition		
Pin Number	Phase Sequence	Wire Color
1	U	Red
2	V	Yellow
3	W	Blue
4	PE	Yellow - green

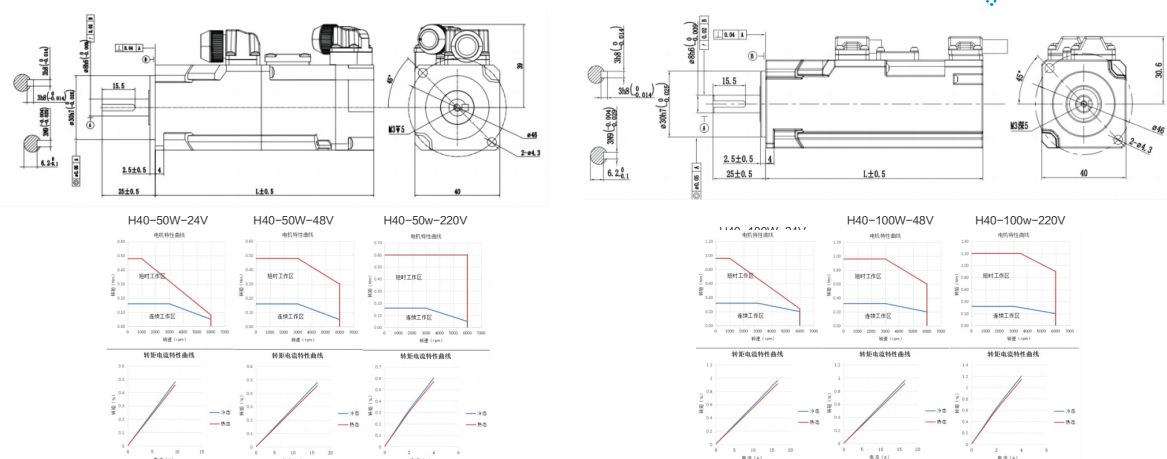
Motor Input Voltage	VDC	24V	48V	220V	24V	48V	220V
Servo Motor Model	/	SY	SY	SY	SY	SY	SY
Flange Specification	mm	□ 40	□ 40	□ 40	□ 40	□ 40	□ 40
Rated Output	W	50	50	50	100	100	100
Number of Pole Pairs	/	5	5	5	5	5	5
Static Torque (Hot State)	Nm	0.16	0.16	0.16	0.32	0.32	0.32
Rated Torque (Hot State)	Nm	0.16	0.16	0.16	0.32	0.32	0.32
Instantaneous Maximum Current	Nm	0.48	0.48	0.48	0.96	0.96	1.20
Rated Current	Arms	3.2	5.5	1	5.5	5.5	1
Instantaneous Maximum Current	Arms	9.6	16.5	4	16.5	16.5	5
Rated Speed	rpm	3000	3000	3000	3000	3000	3000
Maximum Speed (at Maximum Torque)	rpm	1000	3000	6000	1000	3000	3500
Maximum Speed	rpm	4000	6000	6000	4000	6000	6000
Torque Coefficient	Nm/Arms	0.05	0.03	0.16	0.06	0.06	0.32
Moment of Inertia	10 ⁻⁴ kgm ²	0.035	0.035	0.035	0.069	0.069	0.069
Protection Class	/	IP54/65	IP54/65	IP54/65	IP54/65	IP54/65	IP54/65
Insulation Class	/	Class F	Class F	Class F	Class F	Class F	Class F
Wire Resistance	Ohm	1.28	0.50	11.68	0.82	0.79	19.13
Wire Inductance	mh	1.03	0.44	12.29	0.52	0.96	13.30
Wire Back Electromotive Force	V/krpm	3.02	1.95	9.98	3.2	3.95	20.85
Body Length	L	57.5	57.5	57.5	72	72	72

40D Brake Motor

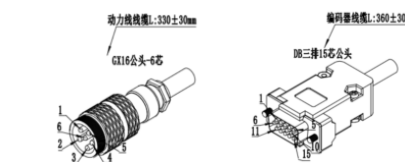
- Maximum speed : 6000rpm
- Protection class : IP54/IP65
- Rated torque : 0.16Nm/0.32Nm
- Rated power : 50W/100W



Product Appearance



Product Specifications



Power Line Definition		
Pin Number	Signal	Wire Color
1	U	Red
2	V	Yellow
3	W	Blue
4	PE	Yellow-green
5	Brake	White
6	Brake	Green

Brake Parameters		
Rated Voltage	VDC	24 ± 10%
Resistance	Ohm	94.4 ± 7%
Excitation Current	ADC	0.25
Holding Torque	Nm	0.32
Pull-in Time	ms	< 50
Release Time	VDC	< 20
Moment of Inertia	10 ⁻⁴ kgm ²	0.00231
Rotational Backlash	°	< 0.5

Definition of 17-bit Absolute Multi-turn Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Blue
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Wiring Definition of Incremental 2500-line Magnetic Encoder

Phase Sequence	Signal	Wire Color
1	GND	Black
2	VCC	Red
3	W+	Gray
4	V+	Orange
5	U+	Blue
6	Z+	Brown
7	B+	White
8	A+	Yellow
9	/	/
10	/	/
11	/	/
12	/	/
13	Z-	Purple
14	B-	Pink
15	A-	Green

Power Cable Definition

Pin Number	Phase Sequence	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Blue
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Motor Input Voltage	VDC	24V	48V	220V	24V	48V	220V
Servo Motor Model	/	SY	SY	SY	SY	SY	SY
Flange Specification	mm	□ 40	□ 40	□ 40	□ 40	□ 40	□ 40
Rated Output	W	50	50	50	100	100	100
Number of Pole Pairs	/	5	5	5	5	5	5
Static Torque (Hot State)	Nm	0.16	0.16	0.16	0.32	0.32	0.32
Rated Torque (Hot State)	Nm	0.16	0.16	0.16	0.32	0.32	0.32
Instantaneous Maximum Current	Nm	0.48	0.48	0.48	0.96	0.96	1.20
Rated Current	Arms	3.2	5.5	1	5.5	5.5	1
Instantaneous Maximum Current	Arms	9.6	16.5	4	16.5	16.5	5
Rated Speed	rpm	3000	3000	3000	3000	3000	3000
Maximum Speed (at Maximum Torque)	rpm	1000	3000	6000	1000	3000	3500
Maximum Speed	rpm	4000	6000	6000	4000	6000	6000
Torque Coefficient	Nm/Arms	0.05	0.03	0.16	0.06	0.06	0.32
Moment of Inertia	10 ⁻⁴ kgm ²	0.035	0.035	0.035	0.069	0.069	0.069
Protection Class	/	IP54/65	IP54/65	IP54/65	IP54/65	IP54/65	IP54/65
Insulation Class	/	Class F	Class F	Class F	Class F	Class F	Class F
Wire Resistance	Ohm	1.28	0.50	11.68	0.82	0.79	19.13
Wire Inductance	mh	1.03	0.44	12.29	0.52	0.96	13.30
Wire Back Electromotive Force	V/krpm	3.02	1.95	9.98	3.2	3.95	20.85
Body Length	L	88.5	88.5	88.5	103	103	103

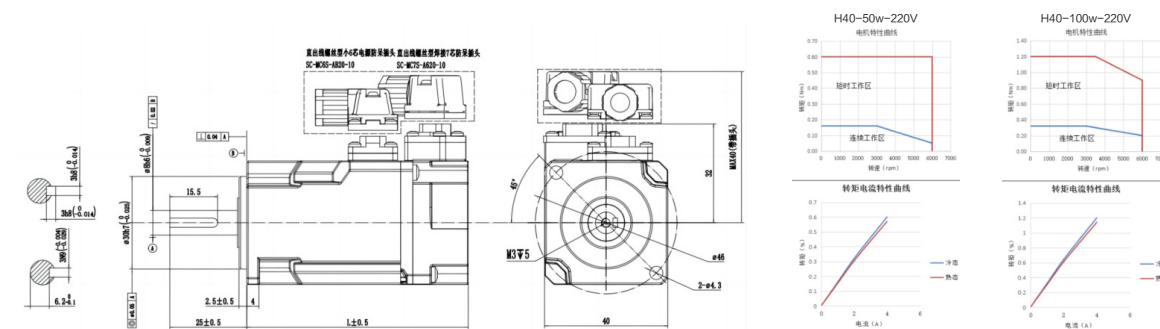
40E motor

50W/100W 220V

- Maximum speed: 6000rpm
- Protection class: IP54/IP65
- Rated torque: 0.16Nm/0.32Nm
- Rated power: 50W/100W



Product Appearance



Product Specifications

Motor Input Voltage	VDC	220V	220V
Servo Motor Model	/	SY	SY
Flange Specification	mm	40	40
Rated Output	W	50	100
Number of Pole Pairs	/	5	5
Static Torque (Hot State)	Nm	0.16	0.32
Rated Torque (Hot State)	Nm	0.16	0.32
Instantaneous Maximum Current	Nm	0.48	1.20
Rated Current	Arms	1	1
Instantaneous Maximum Current	Arms	4	5
Rated Speed	rpm	3000	3000
Maximum Speed (at Maximum Torque)	rpm	6000	3500
Maximum Speed	rpm	6000	6000
Torque Coefficient	Nm/Arms	0.16	0.32
Moment of Inertia	10 ⁻⁴ kgm ²	0.035	0.069
Protection Class	/	IP54/65	IP54/65
Insulation Class	/	Class F	Class F
Wire Resistance	Ohm	11.68	19.13
Wire Inductance	mh	12.29	13.30
Winding back EMF	V/krpm	9.98	20.85
Motor length (L)	L	57.5	72

Definition of 17-bit absolute single-turn magnetic encoder

Pin Number	Signal	Wire Color
1	5V	Red
2	0V	Black
3	SD+	Blue
4	SD-	Yellow
5	VB+	Brown
6	VB-	White
7	PE	Green

Definition of 23-bit absolute multi-turn optical encoder

Pin Number	Signal	Wire Color
1	5V	Red
2	0V	Black
3	SD+	Blue
4	SD-	Yellow
5	VB+	Brown
6	VB-	White
7	PE	Green

Definition of Power Line Lead-out

Pin Number	Definition
1	Phase U
2	Phase V
3	Phase W
4	PE
5	/
6	/

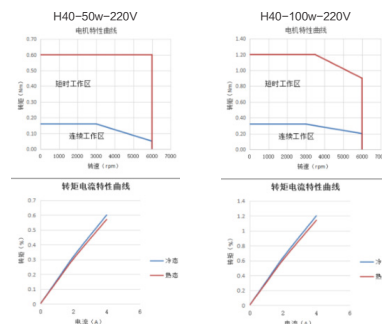
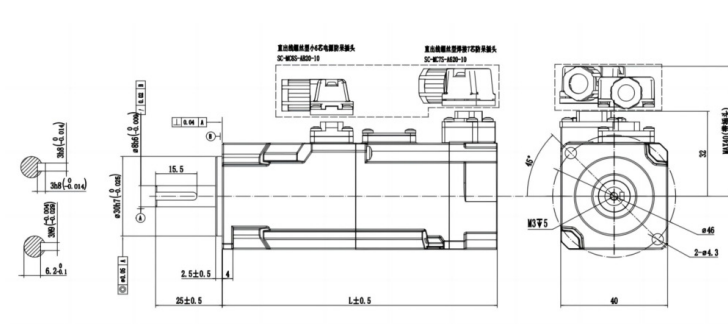
40E Brake motor

50W/100W 220V

- Maximum speed: 6000rpm
- Protection class: IP54/IP65
- Rated torque: 0.16Nm/0.32Nm
- Rated power: 50W/100W



Product Appearance



Product Specifications

Motor Input Voltage	VDC	220V	220V
Servo Motor Model	/	SY	SY
Flange Specification	mm	40	40
Rated Output	W	50	100
Number of Pole Pairs	/	5	5
Static Torque (Hot State)	Nm	0.16	0.32
Rated Torque (Hot State)	Nm	0.16	0.32
Instantaneous Maximum Current	Nm	0.48	1.20
Rated Current	Arms	1	1
Instantaneous Maximum Current	Arms	4	5
Rated Speed	rpm	3000	3000
Maximum Speed (at Maximum Torque)	rpm	6000	3500
Maximum Speed	rpm	6000	6000
Torque Coefficient	Nm/Arms	0.16	0.32
Moment of Inertia	10 ⁻⁴ kgm ²	0.035	0.069
Protection Class	/	IP54/65	IP54/65
Insulation Class	/	Class F	Class F
Wire Resistance	Ohm	11.68	19.13
Wire Inductance	mH	12.29	13.30
Winding back EMF	V/krpm	9.98	20.85
Motor length (L)	L	88.5	103

Brake parameters		
Rated voltage	VDC	24 ± 10%
Resistance	Ohm	94.4 ± 7%
Excitation current	ADC	0.25
Holding torque	Nm	0.32
Pick - up time	ms	< 50
Release time	VDC	< 20
Moment of inertia	10 ⁻⁴ kgm ²	2 0.00231
Rotational backlash	°	< 0.5

Definition of Power Line Lead - out	
Pin Number	Definition
1	Phase U
2	Phase V
3	Phase W
4	PE
5	Brake 1
6	Brake 2

Definition of 17 - bit absolute single - turn magnetic encoder		
Pin Number	Signal	Wire Color
1	5V	Red
2	0V	Black
3	SD+	Blue
4	SD-	Yellow
5	VB+	Brown
6	VB-	White
7	PE	Green

Definition of 23 - bit absolute multi - turn optical encoder		
Pin Number	Signal	Wire Color
1	5V	Red
2	0V	Black
3	SD+	Blue
4	SD-	Yellow
5	VB+	Brown
6	VB-	White
7	PE	Green

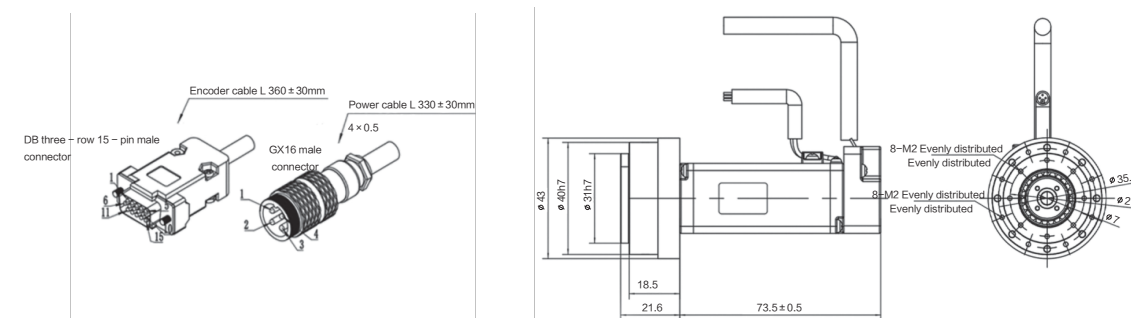
Servo harmonic reduction integrated

33W 48V

- Maximum speed: 60rpm
- Protection class: IP50
- Rated torque: 16Nm
- Rated power: 33W



Product Appearance



Product Specifications

Definition of 17-bit multi-turn absolute magnetic encoder wire color		
Pin Number	Power Line Definition	
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow - green

Model	VDC	8
Reduction Ratio	Ohm	50 100
Rated Torque at 2000r/min Input	Nm	1.8 2.4
Allowable Peak Torque at Start - Stop	Nm	3.3 4.8
Maximum Allowable Average Load Torque	Nm	2.3 3.3
Instantaneous Maximum Allowable Torque	Nm	6.6 9.0
Maximum Allowable Input Speed	r/min	8500
Allowable Average Input Torque	r/min	3500

Rated voltage	V	DC48
Rated power	W	33
Number of pole pairs	/	5
Static torque	Nm	0.105
Rated torque	Nm	0.105
Peak torque	Nm	0.315
Static current	Arms	2
Rated current	Arms	2
Peak current	Arms	6
Rated speed	rpm	3000
Maximum speed	rpm	6000
Torque coefficient	Nm/A	0.0525
Moment of inertia		6.7
Protection class	/	IP50
Insulation class	/	Class F
Winding resistance	Ohm	3.5
Winding inductance	mH	0.92
Winding back EMF	V/krpm	3.5
Total motor length(L)	mm	73.5

Definition of incremental 2500-line magnetic encoder cable		
Phase Sequence	Signal	Wire Color
1	GND	Black
2	Vcc	Red
3	W+	Gray
4	V+	Orange
5	U+	Blue
6	Z+	Brown
7	B+	White
8	A+	Yellow
9	/	/
10	/	/
11	/	/
12	/	/
13	Z-	Purple
14	B-	Pink
15	A-	Green

Definition of 17-bit multi-turn absolute magnetic encoder wire color		
Pin Number	Signal	Wire Color
1	PE	Black
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Definition of 17-bit single-turn absolute magnetic encoder wire color		
Pin Number	Signal	Wire Color
1	PE	Black
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

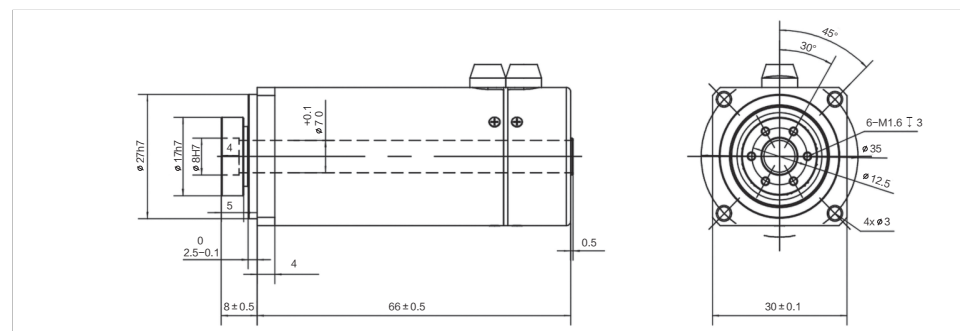
DD30

Ø 30 33W 48V

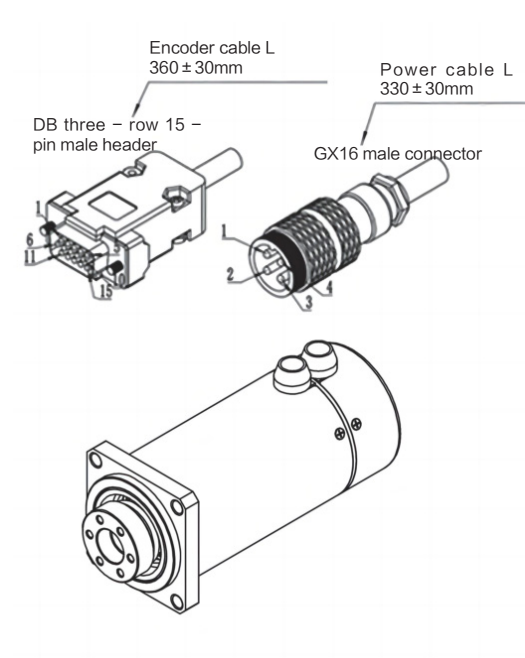
- Maximum speed: 6000rpm
- Protection class: IP50
- Rated torque: 0.315Nm
- Rated power: 33W



Product Appearance



Product Specifications



Rated voltage	V	DC48
Rated power	W	33
Number of pole pairs	/	5
Static torque	Nm	0.105
Rated torque	Nm	0.105
Peak torque	Nm	0.315
Static current	Arms	2
Rated current	Arms	2
Peak current	Arms	6
Rated speed	rpm	3000
Maximum speed	rpm	6000
Torque coefficient	Nm/A	0.0525
Moment of inertia	10 ⁻⁴ kgm ²	6.7
Protection class	/	IP50
Insulation class	/	Class F
Wire resistance	Ω hm	3.4
Wire inductance	mh	0.83
Wire back EMF	V/krpm	3.4
Total motor length(L)	mm	66

Phase Sequence	Power line definition	
1	U	Red
2	V	White
3	W	Black

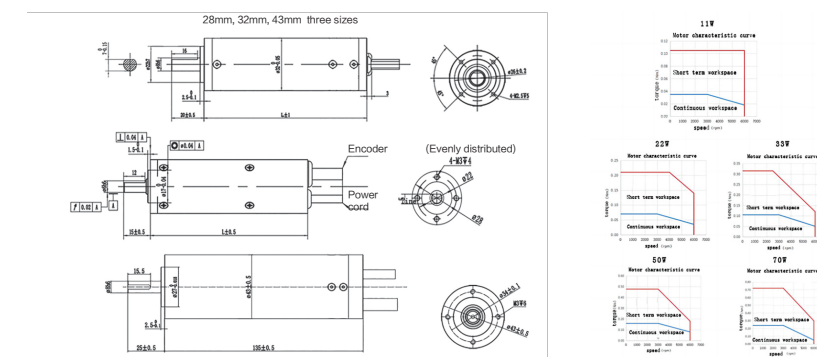
Definition of 17-bit Absolute Value Multi-turn Magnetic Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat shrink tube
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Circular flange servo motor

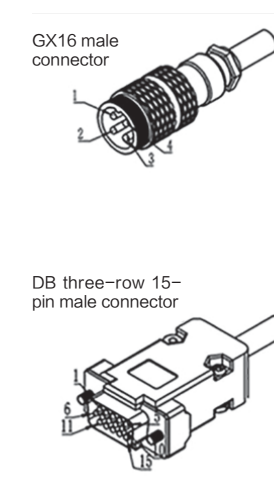
- Maximum speed: 60rpm
- Power range: 11W-200W
- Rated torque : 0.035 ~ 0.5Nm
- Body diameter: Ø28/ Ø32/ Ø43



Product Appearance



Product Specifications



Pin Number	Power Line Definition	
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow - green

Definition of 17-bit Absolute Multi-turn Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tube
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Model	Power W	Rated torque Nm	Rated Speed rpm	Motor Length mm
ModelSYD2800 □ 30M 3005-48V				
SYD2800130M3005-48V	11	0.035	3000	66
SYD2800230M3005-48V	22	0.070	3000	77
SYD2800330M3005-48V	33	0.105	3000	88
SYD2800530M3005-48V	50	0.159	3000	105
SYD3200730M3005-48V	70	0.24	3000	105
SYD4302038M5005-48V	200	0.5	3000	135

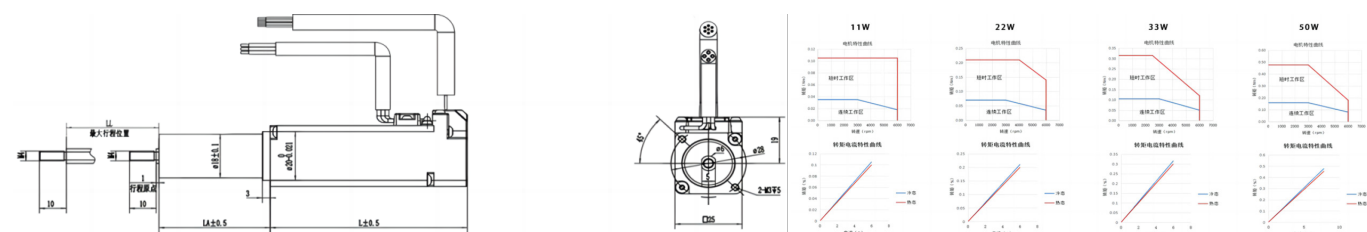
Motor Input Voltage	VDC	48V	48V	48V	48V	48V	48V
Servo Motor Model	/	SY	SY	SY	SY	SY	SY
Flange Specification	mm	□28	□28	□28	□28	□32	□43
Rated Output	W	11	22	33	50	70	200
Number of Pole Pairs	/	5	5	5	5	5	5
Rated Torque	Nm	0.035	0.07	0.105	0.159	0.24	0.5
Instantaneous Maximum Torque	Nm	0.105	0.21	0.315	0.477	0.72	1.5
Rated Current	Arms	2	2	2	2.6	2.5	10.0
Instantaneous Maximum Current	Arms	6	6	6	7.8	7.5	10.0
Rated Speed	rpm	3000	3000	3000	3000	3000	30.0
Maximum Speed (at Maximum Torque)	rpm	6000	6000	6000	6000	6000	3800
Maximum Speed	rpm	6000	6000	6000	6000	6000	6000
Torque Coefficient	Nm/Arms	0.0175	0.035	0.0525	0.06	0.096	0.05
Moment of Inertia	10 ⁻⁴ kgm ²	0.0027	0.0047	0.0067	0.012	0.016	0.175
Protection Class	/	IP50	IP50	IP50	IP50	IP50	IP50
Insulation Class	/	Class F	Class F	Class F	Class F	Class F	Class F
Wire Resistance	Ω hm	1.65	2.45	3.4	1.95	2.4	0.16
Wire Inductance	mh	0.32	0.51	0.83	0.48	0.72	0.17
Wire Back EMF	V/krpm	1.1	2.3	3.4	3.7	4.9	3.00

➤ Servo Electric Linear Actuator

- ◎ Rated linear speed: 50mm/s
- ◎ Repeat positioning accuracy: $\pm 0.005\text{mm}$
- ◎ Horizontal rated thrust: 50N
- ◎ Maximum thrust: 120N/240N/360N/560N



Product Appearance

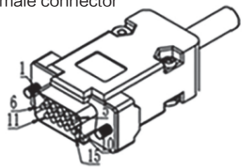


Product Specifications

GX16 male connector



DB three-row 15-pin male connector



Pin Number	Power Line Definition	
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow-green

Definition of 17-bit Absolute Multi-turn Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tube
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

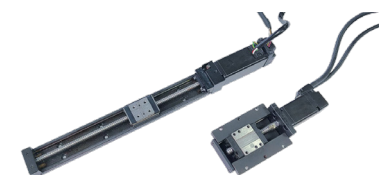
Definition of 17-bit Absolute Multi-turn Encoder		
Pin Number	Signal	Wire Color
1	GND	Black
2	VCC	Red
3	W+	Gray
4	V+	Orange
5	U+	Blue
6	Z+	Brown
7	B+	White
8	A+	Yellow
9	/	/
10	/	/
11	/	/
12	/	/
13	Z-	Purple
14	B-	Pink
15	A-	Green

Model	Power W	Maximum Thrust N	Maximum Stroke mm	Weight kg
SY25D00130M3005-48V-(TG-10)	11	120	10	0.15
SY25D00230M3005-48V-(TG-21)	22	240	21	0.19
SY25D00330M3005-48V-(TG-32)	33	360	32	0.23
SY25D00530M3005-48V-(TG-49)	50	560	49	0.27

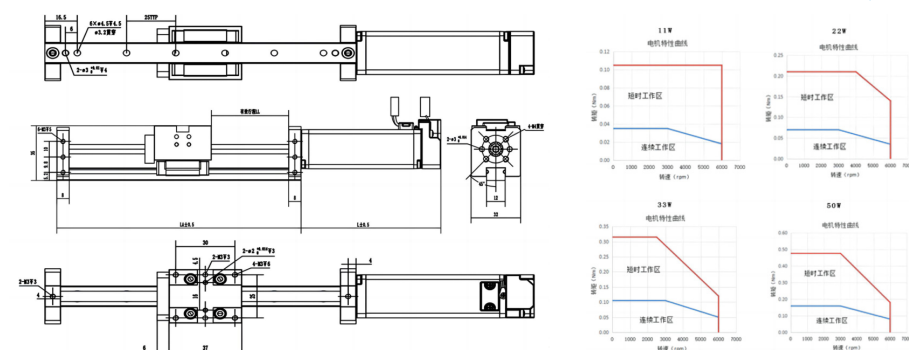
Motor Input Voltage	VDC	48V	48V	48V	48V
Servo Motor Model	/	SY	SY	SY	SY
Flange Specification	mm	□ 25	□ 25	□ 25	□ 25
Rated Output	W	11	22	33	50
Number of Pole Pairs	/	5	5	5	5
Rated Torque (Hot State)	Nm	0.035	0.07	0.105	0.159
Instantaneous Maximum Torque	Nm	0.105	0.21	0.315	0.477
Rated Current	Arms	2	2	2	2.6
Instantaneous Maximum Current	Arms	6	6	6	7.8
Rated Speed	rpm	3000	3000	3000	3000
Maximum Speed (at Maximum Torque)	rpm	6000	6000	6000	6000
Maximum Speed	rpm	6000	6000	6000	6000
Torque Coefficient	Nm/Arms	0.0175	0.035	0.0525	0.06
Moment of Inertia	10 ⁻⁴ kgm ²	0.0027	0.0047	0.0067	0.012
Protection Class	/	IP50	IP50	IP50	IP50
Insulation Class	/	Class F	Class F	Class F	Class F
Wire Resistance	Ω/m	1.65	2.45	3.4	1.95
Wire Inductance	mH	0.32	0.51	0.83	0.48
Wire Back EMF	V/krpm	1.1	2.3	3.4	3.7

➤ Servo Slide Module

- ◎ Linear Speed: 50-200mm/s
- ◎ Repeat Positioning Accuracy: $\pm 0.005\text{mm}$
- ◎ Power Range: 11W-50W
- ◎ Encoder Precision: 17bit

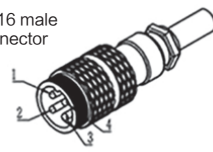


Product Appearance

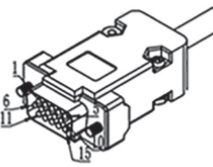


Product Specifications

GX16 male connector



DB three-row 15-pin male connector



Pin Number	Power Line Definition	
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow-green

Definition of 17-bit Absolute Multi-turn Encoder		
Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tube
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Definition of 17-bit Absolute Multi-turn Encoder		
Pin Number	Signal	Wire Color
1	GND	Black
2	VCC	Red
3	W+	Gray
4	V+	Orange
5	U+	Blue
6	Z+	Brown
7	B+	White
8	A+	Yellow
9	/	/
10	/	/
11	/	/
12	/	/
13	Z-	Purple
14	B-	Pink
15	A-	Green

Model	Power W	Maximum Horizontal Load kg	Maximum Vertical Load kg	Maximum Stroke mm
SY25D00130 □ 005-48V-(MZ-LL)	11	15	10	LL
SY25D00230 □ 005-48V-(MZ-LL)	22	30	25	LL
SY25D00330 □ 005-48V-(MZ-LL)	33	40	35	LL
SY25D00530 □ 005-48V-(MZ-LL)	50	65	60	LL

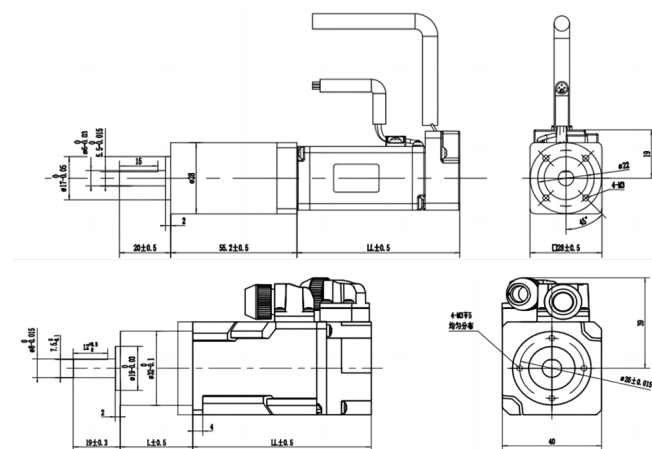
Motor Input Voltage	VDC	48V	48V	48V	48V
Servo Motor Model	/	SY	SY	SY	SY
Flange Specification	mm	□ 25	□ 25	□ 25	□ 25
Rated Output	W	11	22	33	50
Number of Pole Pairs	/	5	5	5	5
Rated Torque (Hot State)	Nm	0.035	0.07	0.105	0.159
Instantaneous Maximum Torque	Nm	0.105	0.21	0.315	0.477
Rated Current	Arms	2	2	2	2.6
Instantaneous Maximum Current	Arms	6	6	6	7.8
Rated Speed	rpm	3000	3000	3000	3000
Maximum Speed (at Maximum Torque)	rpm	6000	6000	6000	6000
Maximum Speed	rpm	6000	6000	6000	6000
Torque Coefficient	Nm/Arms	0.0175	0.035	0.0525	0.06
Moment of Inertia	10 ⁻⁴ kgm ²	0.0027	0.0047	0.0067	0.012
Protection Class	/	IP50	IP50	IP50	IP50
Insulation Class	/	Class F	Class F	Class F	Class F
Wire Resistance	Ω/m	1.65	2.45	3.4	1.95
Wire Inductance	mH	0.32	0.51	0.83	0.48
Wire Back EMF	V/krpm	1.1	2.3	3.4	3.7

➤ Servo gear reduction integrated machine

- ◎ Power range: 11W–50W
- ◎ Protection class: IP50
- ◎ Maximum output torque: 6Nm
- ◎ Maximum output shaft speed: 1500rpm

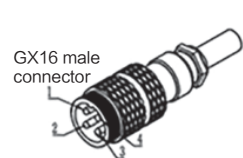


Product Appearance

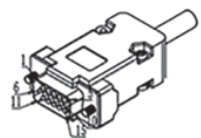


Reducer parameters				
Reduction ratio	Output torque Nm	Maximum torque Nm	Transmission efficiency %	Reducer length m
3.5	2.0	3.5	90	29.3
4.0	2.0	3.5	90	29.3
4.6	2.0	3.5	90	29.3
5.5	2.0	3.5	90	29.3
12.76	2.5	3.5	81	37.1
16.0	2.5	3.5	81	37.1
18.4	2.5	3.5	81	37.1
22.0	2.5	3.5	81	37.1
25.3	2.5	3.5	81	37.1
30.25	2.5	3.5	81	37.1
45.55	3.0	4.5	73	42.4
51.02	3.0	4.5	73	42.4
57.14	3.0	4.5	73	42.4
64.0	3.0	4.5	73	42.4

Product Specifications



DB three-row 15-pin male connector



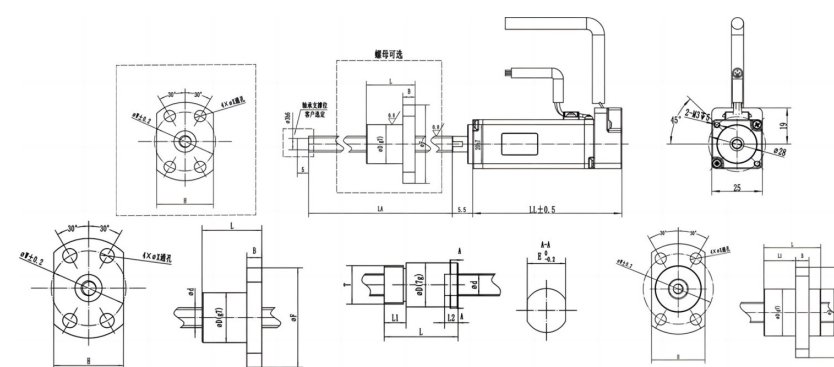
Pin Number	Power cord definition		Rated voltage	V	DC48	DC48	Reducer parameters								
1	U	Red	Rated power	W	11W/22W/33W/50W	50W/100W	Reduction ratio	Output torque Nm	Maximu m torque Nm	Transmission efficiency %	Reducer length m				
2	V	Yellow/White	Pole pairs	/	5	5	3.705	0.5	1.5	80	35				
3	W	Blue/Black	Static torque	Nm	0.035/0.070/0.105/0.159	0.16/0.32	5.1818	0.5	1.5	80	35				
4	PE	Yellow-green	Rated torque	Nm	0.035/0.070/0.105/0.159	0.16/0.32	13.733	1.0	3.0	70	44.4				
Definition of 17 – bit absolute multi – turn magnetic encoder							19.203	1.0	3.0	70	44.4				
							Peak torque	Nm	0.105/0.210/0.315/0.478	0.48/0.96	26.851	1.0	3.0	70	44.4
							Static current	Arms	2/2/2/2.6	5.5/5.0	50.895	1.5	4.5	60	55.2
							Rated current	Arms	2/2/2/2.6	5.5/5.0	71.165	1.5	4.5	60	55.2
							Peak current	Arms	6/6/6/7.8	16.5/16.5	99.506	1.5	4.5	60	55.2
Pin Number	Signal	Wire Color	Rated speed	rpm	3000	3000	139.185	1.5	4.5	60	55.2				
1	PE	Heat shrink tube	Maximum speed	rpm	6000	6000	188.611	2.0	6.0	50	62.8				
2	5V	Red	Torque coefficient	Nm/Arms	0.0175/0.035/0.0525/0.06	0.03/0.064	263.728	2.0	6.0	50	62.8				
3	0V	Black	Moment of inertia	10 ⁻⁷ kgm ²	2.7/4.7/6.7/11.7	35/69	368.763	2.0	6.0	50	62.8				
4	SD+	Blue/Green	Protection class	/	IP50	IP54/65	515.630	2.0	6.0	50	62.8				
5	SD–	Yellow	Insulation class	/	Class F	Class F	720.989	2.0	6.0	50	62.8				
6	VB+	Brown	Wire resistance	0 hm	1.7/2.6/3.5/5.3	0.5/0.79									
7	VB–	White	Wire inductance	mH	0.26/0.49/0.73/1.1	0.44/0.96									
			Wire back EMF	V/krpm	1.2/2.4/3.5/4.8	1.95/3.95									
			Total body length(L)	mm	51.5/62.5/73.5/90.5	57.5/72									

➤ Servo screw integrated machine

- ◎ Power range: 5W–100W
- ◎ Repeat positioning accuracy: ±0.01mm
- ◎ Optional lead: 01/02/04/06
- ◎ Screw accuracy class: C5/C3



Product Appearance

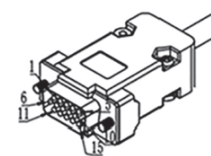


MIF(MUF) Lead Screw Nut Selection Table									
Model	d	D	F	L	B	W	H	X	
MIF0401	4	10	20	12	3	15	14	2.9	
MIF0601	6	12	24	15	3.5	18	16	3.4	
MIF0602	6	12	24	15	3.5	18	16	3.4	
MIF0802	8	14	27	16	4	21	18	3.4	
MIF0804	8	16	29	26	4	23	20	3.4	
MIA Lead Screw Nut Selection Table									
Model	d	D	L	L1	L2	E	T		
MIA0401	4	10	15	5	–	–	M8×0.75		
MIA0601	6	12	16	5	2	10	M10×1		
MIA0602	6	12	20	6	–	–	M10×1		
MIA0801	8	16	22	8	–	–	M14×1		
MIA0802	8	16	27	8	–	–	M14×1		
MEF Lead Screw Nut Selection Table									
Model	d	D	F	L	L1	B	W	H	X
MIF0401	6	14	27	17	9.5	4	21	16	3.4
MIF0601	8	18	31	21.5	13	4	25	20	3.4
MIF0602	8	18	31	24	13	4	25	20	3.4

Product Specifications



DB three-row 15-pin male connector



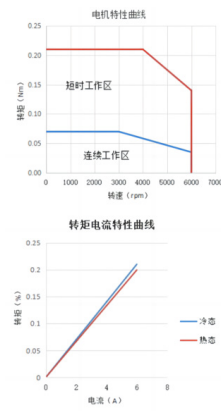
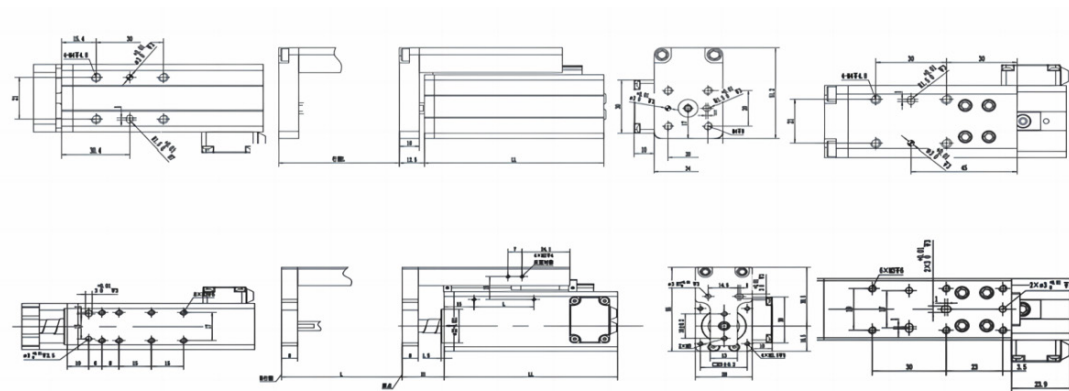
Pin Number	Power cord definition		Rated Voltage	V	DC48	DC48	DC48
1	U	Red	Rated Power	W	11W/22W/33W/50W	50W/100W	5W
2	V	White/Yellow	Pole Pairs	/	5	5	2
3	W	Black/Blue	Static Torque	Nm	0.035/0.070/0.105/0.159	0.16/0.32	0.016
4	PE	Yellow – Green	Peak Torque	Nm	0.105/0.210/0.315/0.478	0.48/0.96	0.048
17-bit absolute encoder definition							
Pin Number	Signal	Wire Color	Static Current	Arms	2/2/2/2.6	5.5/5.0	1.5
1	PE	heat shrink tube	Rated Current	Arms	2/2/2/2.6	5.5/5.0	1.5
2	5V	Red	Peak Current	Arms	6/6/6/7.8	16.5/16.5	4.5
3	0V	Black	Rated Speed	rpm	3000	3000	3000
4	SD+	Blue/Green	Maximum Speed	rpm	6000	6000	6000
5	SD-	Yellow	Torque Coefficient	Nm/Arms	0.0175/0.035/0.0525/0.06	0.03/0.064	0.011
6	VB+	Brown	Moment of Inertia	10 ⁻⁷ kgm ²	2.7/4.7/6.7/11.7	35/69	0.7
7	VB-	White	Protection Class	/	IP50	IP50	IP50
				Insulation Class	/	Class F	Class F
				Winding Resistance	0 hm	1.7/2.6/3.5/5.3	0.5/0.79
				Winding Inductance	mH	0.26/0.49/0.73/1.1	0.44/0.96
				Winding Back EMF	V/krpm	1.2/2.4/3.5/4.8	1.95/3.95
				Total Length of Body(L)	mm	51.5/62.5/73.5/90.5	57.5/72

➤ Servo Electric Cylinder

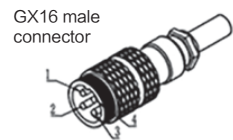
- ◎ Power range: 11W–50W
- ◎ Protection class: IP50
- ◎ Maximum output torque: 6Nm
- ◎ Maximum output shaft speed: 1500rpm



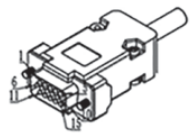
Product Appearance



Product Specifications



DB three-row 15-pin male connector



Pin Number	Power Line Definition	
1	U	Red
2	V	White
3	W	Black
4	PE	Yellow-green

Definition of 17-bit Absolute Multi-turn Encoder

Pin Number	Signal	Wire Color
1	PE	Heat Shrink Tubing
2	5V	Red
3	0V	Black
4	SD+	Green
5	SD-	Yellow
6	VB+	Brown
7	VB-	White

Model	SY □ L02 □ M5 □ 0D	Power W	Vertical Load N	Axial Thrust N	Stroke L mm	Body LL mm
SY36L0230M5020D		22	100	150	30	90
SY36L0250M5030D		33	175	225	50	110
SY36L0275M5050D		50	280	330	75	135
SY28L0230M5020D		22	100	150	30	87
SY28L0250M5030D		33	175	225	50	107
SY28L0275M5050D		50	280	330	75	132

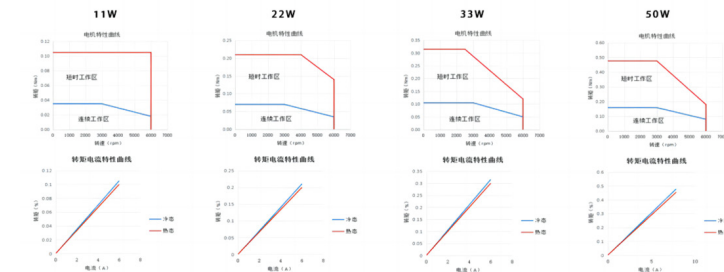
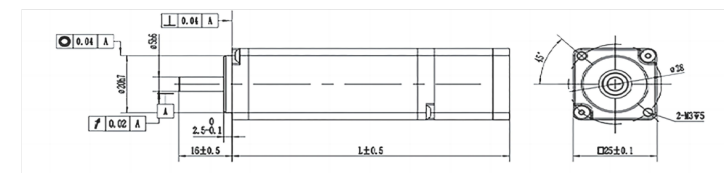
Motor input voltage	VDC	48V	48V	48V	48V	48V	48V
Servo motor model	/	SY	SY	SY	SY	SY	SY
Flange specification	mm	SY28L	SY28L	SY28L	SY36L	SY36L	SY36L
Rated output	W	22	33	50	22	33	50
Number of pole pairs	/	5	5	5	5	5	5
Rated torque (hot state)	Nm	0.07	0.105	0.159	0.07	0.105	0.159
Instantaneous maximum torque	Nm	0.21	0.315	0.477	0.21	0.315	0.477
Rated current	Arms	2	2	2.6	2	2	2.6
Instantaneous maximum current	Arms	6	6	7.8	6	6	7.8
Rated speed	rpm	3000	3000	3000	3000	3000	3000
Maximum speed (at maximum torque)	rpm	6000	6000	6000	6000	6000	6000
Maximum speed	rpm	6000	6000	6000	6000	6000	6000
Torque coefficient	Nm/Arms	0.035	0.0525	0.06	0.035	0.0525	0.06
Moment of inertia	10 ⁻⁴ kgm ²	0.0047	0.0067	0.012	0.0047	0.0067	0.012
Protection class	/	IP50	IP50	IP50	IP50	IP50	IP50
Insulation class	/	Class F	Class F	Class F	Class F	Class F	Class F
Wire resistance	Ω/m	2.45	3.4	1.95	2.45	3.4	1.95
Wire inductance	mH	0.51	0.83	0.48	0.51	0.83	0.48
Wire back electromotive force	V/krpm	2.3	3.4	3.7	2.3	3.4	3.7

➤ Servo drive all – in – one machine

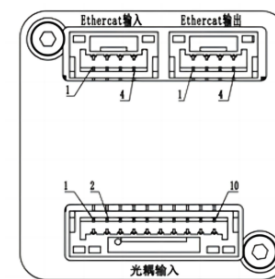
- ◎ Power range: 5W–100W
- ◎ Communication type: EtherCAT
- ◎ Maximum torque: 0.477Nm
- ◎ Maximum speed: 4000rpm



Product Appearance



Product Specifications



EtherCAT Input/Output Pin Definition	
1	T-
2	T+
3	R-
4	R+

Model	SY25D00 □ 30M 3005-24V-(QK)	Power W	Rated torque Nm	L mm	Weight kg
SY25D00130M3005-24V-(QK)		11	0.035	62	0.12
SY25D00230M3005-24V-(QK)		22	0.07	73	0.14
SY25D00330M3005-24V-(QK)		33	0.105	84	0.18
SY25D00530M3005-24V-(QK)		50	0.159	101	0.23

Optocoupler input pin definition	
1	Power negative
2	Power positive
3	Power negative
4	Power positive
5	Power positive
6	Power positive
7	Output
8	Input 1
9	Input 2
10	Input common terminal

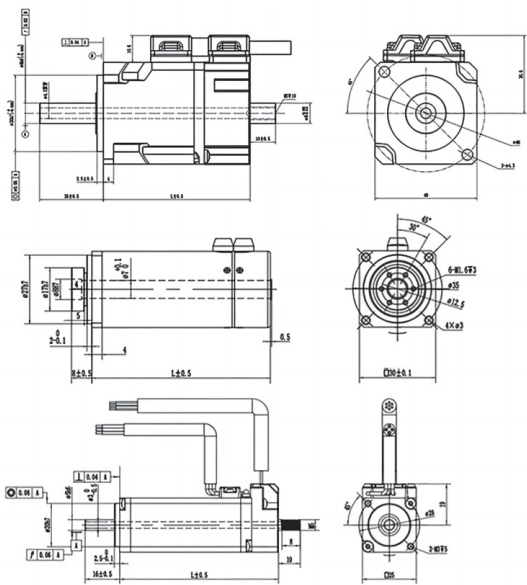
Motor input voltage	VDC	24V	24V	24V	24V
Servo motor model	/	SY	SY	SY	SY
Flange specification	mm	□ 25	□ 25	□ 25	□ 25
Rated output	W	11	22	33	50
Number of pole pairs	/	5	5	5	5
Rated torque (hot state)	Nm	0.035	0.07	0.105	0.159
Instantaneous maximum torque	Nm	0.105	0.21	0.315	0.477
Rated current	Arms	2	2	2	2.6
Instantaneous maximum current	Arms	6	6	6	7.8
Rated speed	rpm	3000	3000	3000	3000
Torque coefficient	Nm/Arms	0.0175	0.035	0.0525	0.06
Moment of inertia	10 ⁻⁴ kgm ²	0.0027	0.0047	0.0067	0.012
Protection class	/	IP50	IP50	IP50	IP50
Insulation class	/	Class F	Class F	Class F	Class F
Wire resistance	Ω/m	1.65	2.45	3.4	1.95
Wire inductance	mH	0.32	0.51	0.83	0.48
Wire back electromotive force	V/krpm	1.1	2.3	3.4	3.7

➤ Servo Hollow Shaft Motor

- ◎ Rated Speed : 3000rpm
- ◎ Power Range: 11W~100W
- ◎ Rated Torque: 0.035 ~ 0.32Nm
- ◎ Protection Class: IP50



Product Appearance



Model SY40D00 □ 30M 5015-48V~(HD2-AGH)	Power W	Rated Torque Nm	Rated Speed rpm	Length mm
SY40D00 □ 30M5015-48V~(HD2-AGH)	50	0.16	3000	57.5
SY40D00 □ 30M5015-48V~(HD2-AGH)	100	0.32	3000	72

ModelSY25D00 □ 30C2005-48V~(HD)	Power W	Rated Torque Nm	Rated Speed rpm	Length mm
SY25D00 □ 30C2005-48V~(HD)	11	0.035	3000	51.5
SY25D00 □ 30C2005-48V~(HD)	22	0.070	3000	62.5
SY25D00 □ 30C2005-48V~(HD)	33	0.105	3000	73.5
SY25D00 □ 30C2005-48V~(HD)	50	0.159	3000	90.5

ModelSYD3000 □ 30M5005-48V~(ZK)	Power W	Rated Torque Nm	Rated Speed rpm	Length mm
SYD3000 □ 30M5005-48V~(ZK)	11	0.035	3000	48
SYD3000 □ 30M5005-48V~(ZK)	22	0.070	3000	50
SYD3000 □ 30M5005-48V~(ZK)	33	0.105	3000	70
SYD3000 □ 30M5005-48V~(ZK)	50	0.159	3000	53

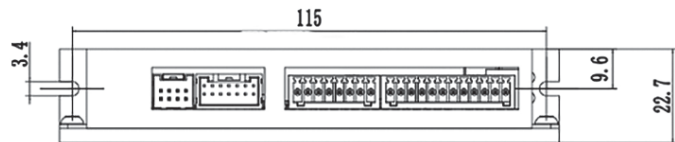
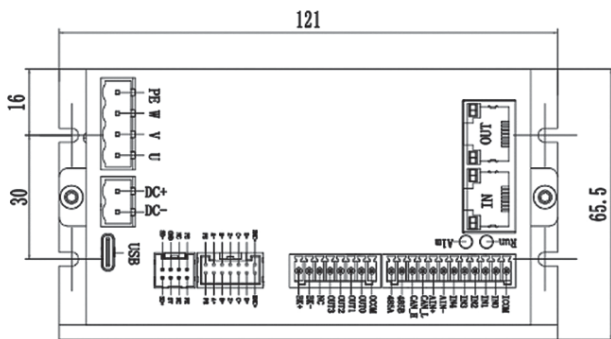
Motor Input Voltage	VDC	48V	48V	48V	48V	24V	48V	220V	24V	48V	220V
Servo Motor Model	/	SY	SY	SY	SY	SY	SY	SY	SY	SY	SY
Flange Specification	mm	□ 25	□ 25	□ 25	□ 25	□ 40	□ 40	□ 40	□ 40	□ 40	□ 40
Rated Output	W	11	22	33	50	50	50	50	100	100	100
Number of Pole Pairs	/	5	5	5	5	5	5	5	5	5	5
Rated Torque	Nm	0.035	0.07	0.105	0.159	0.16	0.16	0.16	0.32	0.32	0.32
Instantaneous Maximum Torque	Nm	0.105	0.21	0.315	0.477	0.48	0.48	0.48	0.96	0.96	1.20
Rated Current	Arms	2	2	2	2.6	3.2	5.5	1	5.5	5.5	1
Instantaneous Maximum Current	Arms	6	6	6	7.8	9.6	16.5	4	16.5	16.5	5
Rated Speed	rpm	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000
Maximum Speed (at Maximum Torque)	rpm	6000	6000	6000	6000	1000	3000	6000	1000	3000	3500
Maximum Speed	rpm	6000	6000	6000	6000	4000	6000	6000	4000	6000	6000
Torque Coefficient	Nm/Arms	0.0175	0.035	0.0525	0.06	0.05	0.03	0.16	0.06	0.06	0.32
Moment of Inertia	10 ⁻⁴ kgm ²	0.0027	0.0047	0.0067	0.012	0.035	0.035	0.035	0.069	0.069	0.069
Protection Class	/	IP50	IP50	IP50	IP50	IP50	IP50	IP50	IP50	IP50	IP50
Insulation Class	/	Class F	Class F	Class F	Class F	Class F	Class F	Class F	Class F	Class F	Class F
Wire Resistance	Ω hm	1.65	2.45	3.4	1.95	1.28	0.50	11.68	0.82	0.79	19.13
Wire Inductance	mH	0.32	0.51	0.83	0.48	1.03	0.44	12.29	0.52	0.96	13.30
Wire Back Electromotive Force	V/krpm	1.1	2.3	3.4	3.7	3.02	1.95	9.98	3.2	3.95	20.85

➤ EtherCAT Bus – type Drive(YKB300)

- ◎ Multiple control modes: position mode, speed mode, torque mode
- ◎ Lightweight:121×65.5×22.7
- ◎ Flexible installation: front and side mounting holes
- ◎ Communication :EtherCAT



Product Appearance



Product Specifications

Product Series	YKB300: Represents YKB300 Series
Number of Axes	01: Indicates Single Axis
Rated Input Voltage Level	A: IndicatesDC48V
Single – axis Output Current	001:Corresponds1A
	005:Corresponds5A
	0010:Corresponds10A
	0015:Corresponds15A
Upper – level Control Mode	E:IndicatesEtherCAT
STO Function	A: Without STO
	S: WithSTO
Special Function	0: No Special Function
External Structure	0: Default External Structure
Operating Temperature Grade	0: Indicates0 ~ 50℃
Encoder Feedback	A: Incremental Encoder
	B: Rotary Absolute Encoder (Tamagawa)

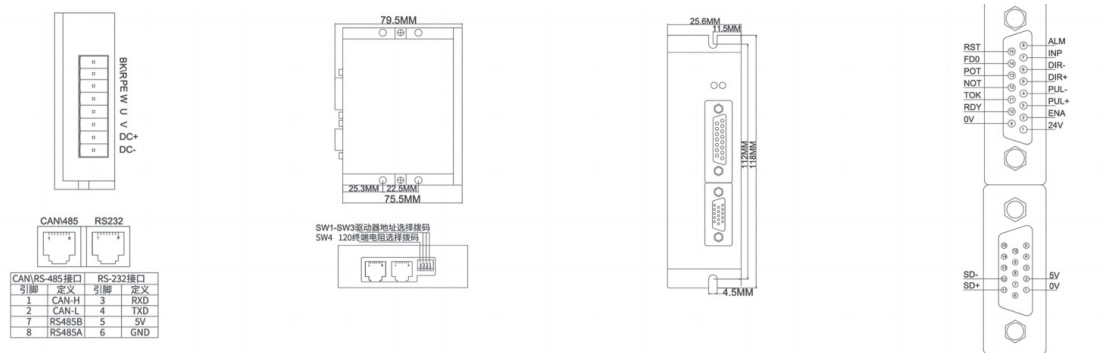
Output Power	Rated Power (W)	≤ 400W
	Rated Current (A)	≤ 15A
	Instantaneous Peak Current (A)	45A
Input Power	Duration of Instantaneous Peak Current (S)	3S
	Rated Input Voltage (VDC)	24 ~ 48V
	Maximum Input Voltage (VDC)	48V
Logic Power Supply (When Independent of Power Circuit)	Rated Input Voltage (VDC)	24V
	Maximum Input Voltage (VDC)	24V
Control Mode	Bus	Ethercat
	Pulse	/
	Debugging	Ethercat/USB
DigitalI/O	Input	5 Isolated Channels, Configurable
	Output	4 Isolated Channels
Analog Quantity	Input	± 10V, 1 Channel, 12 – bit Resolution
Motor Feedback		Rotary Absolute Encoder(Tamagawa)
Status Display		Dual – channel LEDs: Red, Green
Motor Control		Position, Speed, Torque Three – loop Control; Current Loop/Position Loop/Speed Loop 62.5 μ s
Fault Protection		Short Circuit, Overcurrent, Overtemperature, Overvoltage, Undervoltage, Overspeed, Encoder Communication Abnormality, etc.
Dimension		121 × 72 × 23
Weight		200g

➤ Pulse – type Drive(ZKSY830ABS)

- ◎ Multiple control modes: position mode, speed mode, torque mode
- ◎ Lightweight: 118 × 75.5 × 27.5
- ◎ Flexible installation: mounting holes on the front and side
- ◎ Communication type: pulse 485



Product Appearance



Product Specifications

Output Power	Rated Power(W)	≤ 400W	Serial Number	Mark	Wiring Function	Wiring Description				
	Rated Current (A)	≤ 20A			1	DC (-)	Input Power Supply(-)	DC 24V~72V		
	Instantaneous Peak Current(A)	60A			2	DC (+)	Input Power Supply(+)			
	Duration of Instantaneous Peak Current(S)	3S			3	U	Motor Power Line Phase U		Must Be Connected to the Motor According to the Mark	
					4	V	Motor Power Line Phase V			
5					W	Motor Power Line Phase W				
Input Power	Rated Input Voltage(VDC)	24 ~ 48V	6	PE	Motor Ground Wire	7	BK/R (2 Choices 1)	Braking Resistor (Enabled When Discharge Voltage Parameter Threshold = 85V)	External Braking Resistor, Energy Consumption Discharge (Choose According to Manufacturer's Recommendation)	
	Maximum Input Voltage(VDC)	48V			8					Brake Interface (Enabled When Discharge Voltage Parameter Threshold = 85V)
Logic Power Supply (When Independent of Power Circuit)	Rated Input Voltage (VDC)	24V								
	Maximum Input Voltage (VDC)	24V								
Control Mode	Bus	/								
	Pulse	Modbus(RS485)								
	Debugging	RS232								
Motor Feedback Input		Rotary Absolute Encoder (Tamagawa)								
Status Display		Dual – channel LED: RED/GREEN								
Motor Control		Position, Speed, Torque Three – loop Control; Current Loop/Position Loop/Speed Loop 62.5 μ s								
Fault Protection		Short Circuit, Overcurrent, Overtemperature, Overvoltage, Undervoltage, Overspeed, Encoder Communication Abnormality, etc.								
Dimension		118mm × 75.5mm × 27.5mm								
Weight		250g								

1		EX_24V	External Power Supply 24V Input Positive Pole (For I/O and DO Power Supply)
2		Servo Enable Input	Signal Level 24V, Used for Enabling/Disabling the Servo Motor. At 0V, the Driver Will Cut Off the Motor Power Supply, Leaving the Motor in a Free State and Not Responding to Pulses. Enabled When +24V or Disconnected.
3		Pulse CommandPULSE(+)	Pulse Signal: Rising Edge of Pulse is Valid; High Level: 4 ~ 5V, Low Level: 0 ~ 0.5V; Pulse Width Should Be Greater Than 1.6 μ s
4		Pulse Command PULSE(-)	
5		Direction CommandDIR+	Direction Signal Should Be Established at Least 5 μ s Before Pulse Signal; High Level: 4 ~ 5V, Low Level: 0 ~ 0.5V
6		Direction CommandDIR-	
7		Position Completion Output	Position Completion Output 0 or 0V Output (Open Collector)
8		Servo Alarm Output	Alarm Output (Open Collector)
9		EX_GND/0V	Power Supply Negative Pole 0V Input
10		Servo Ready Output	Drive Power – on Completion Output (Open Collector)
11		Torque Arrival Output	Output After Torque Reaches Set Value (Open Collector)
12		Reverse Direction Inhibition Input	Reverse Rotation Stop, Forward Rotation Can Run (Open Collector)
13		Forward Direction Inhibition Input	Forward Rotation Stop, Reverse Rotation Can Run (Open Collector)
14		Home Start Input	Electric Operation to Absolute Zero Point (0V Valid)

